Contents

1 WORKFLOW: regional stress → on-fault traction VTU

Target artefact (this doc explains exactly how it is produced):

```
stress/results/safs_fault_box_nwcut_500m_lcfar3000_z0embed_hz1671_psi37_az351/
    safs_fault_box_nwcut_500m_lcfar3000_z0embed_hz1671_psi37_az351_fault_stress.vtu
    └ sibling: ..._bulk_stress.vtu, ..._summary.json
```

Written 2026-05-28. This is a **per-run map** for the `hz1671_psi37_az351` fault-stress projection on the 500 m `z0embed` mesh. It is the run-specific companion to `stress/README.md` (the general pipeline reference) — read the README §Workflow for the multi-variant / batch view; read this for *what the filename tags mean*, *the exact command that made this file*, and *the numbers that came out*.

The whole stress pipeline is **pure Python** (`meshio` + `numpy`) and produces only visualization / IC-setup artefacts. No C++ runtime is involved in making this VTU (the C++ `stress_safs.h5` sidecar is a separate, optional branch — see README §C).

1.1 TL;DR — the one command that produced the target file

```
conda activate pythonenv
cd stress/code

python project_to_fault_stress.py \
    ../../meshing/results/vtu/safs_fault_box_nwcut_500m_lcfar3000_z0embed_fault.vtu \
    safs_fault_box_nwcut_500m_lcfar3000_z0embed_hz1671_psi37_az351 \
    --write-bulk \
    --SHmax-az 351
```

- **arg 1** = input fault VTU (from the meshing pipeline, see Stage 0).
- **arg 2** = `mesh_base` — the output stem *and* the output sub-directory name.
- `--write-bulk` → also emit the `_bulk_stress.vtu` (σ^0 on every tet centroid).
- `--SHmax-az 351` is the **only** parameter override. Everything else (`SHmax/Shmin/Sv/P_p` magnitudes and the `--strike-hint-az 314`) is left at the built-in Hickman & Zoback 1671 m defaults (`DEFAULTS = _hz_demo_safod_defaults()`, `project_to_fault_stress.py:1480`).

  Why the output dir is the full base name (not `500m_lcfar3000/`): `_extract_lc_tag()` matches `_<N>m` followed by **at most one** variant suffix (`r"_(\d+m(?:_[A-Za-z0-9]+)?)$"`). The stem `..._500m_lcfar3000_z0embed_hz1671_psi37_az351` has many trailing segments and ends in `az351`, so the regex misses and the function **falls back to the full `mesh_base`**. That fallback is what created the long-named result directory.

1.2 Filename tag decode

`safs_fault_box_nwcut_500m_lcfar3000_z0embed _hz1671 _psi37 _az351`

Segment	Meaning	Source
`safs_fault_box_nwcut`	SAFS box mesh with the NW hard-cut applied	meshing pipeline
`500m`	500 m near-fault target resolution	meshing
`lcfar3000`	far-field background size `lc = 3 km`	meshing
`z0embed`	fault cut **exactly at z = 0** (`run_z0cut_meshing.py`)	meshing Stage 5
`hz1671`	**H**ickman & **Z**oback (2004) SAFOD magnitudes at **1671 m** depth: `SHmax=113, Shmin=49, Sv=45, P_p=16` MPa	`…_projection.py:328,388`
`psi37`	**$\Psi \approx 37^\circ$** — angle between SHmax and the SAF strike (`|351 − 314| = 37`). The *intermediate/shallow* value from H&Z's depth table, vs the deep $\Psi = 69^\circ$ used in `demo_safod()`	`…_projection.py:379`
`az351`	**SHmax azimuth = 351**$^\circ$ cw-from-N (= N9°W). Chosen so that, against the SAF strike hint 314°, $\Psi = 37^\circ$	`--SHmax-az 351`

The tags are an operator-supplied label baked into `mesh_base`; the script does **not** parse them. They are only descriptive — the actual numeric parameters are the ones echoed in `_summary.json["params"]`.

1.3 Pipeline at a glance

```
meshing/results/vtu/…_z0embed_fault.vtu  ┐  (triangulated SAF surface)
                          (+ _bulk.vtu)  ┤
                                         │
   H&Z 1671 m regional params  ──────────┤
   SHmax=113 Shmin=49 Sv=45 P_p=16       │
   SHmax_az=351  fault_strike_hint=314   │
                                         ▼
           project_to_fault_stress.py  (single-file mode)
                                         │
   (1) build_bulk_stress_tensor(σ^0)  3×3 Cauchy in (E,N,Up), compression POSITIVE
   (2) triangle_geometry              per-tri centroid + unit normal
   (3) build_fault_basis             Tandem (s = up×n̂, d = s×n̂, n̂), strike→314° hint
  (4) resolve_traction(σ^0,n̂,s,d,P_p) σ_n, σ_n_eff, τ_strike, τ_dip, |τ|, rake, μ_app
   (5) write_fault_vtu / write_bulk_vtu / write_summary_json
                                         ▼
   results/<full mesh_base>/…_fault_stress.vtu  + _bulk_stress.vtu + _summary.json
```

1.4 Stage 0 – input mesh (from the meshing pipeline)

The projector consumes the **ParaView VTU split** of the volume `.msh` (produced by `msh_to_vtu.py`, PIPELINE Stage 7) — two siblings under `meshing/results/vtu/`:

- `safs_fault_box_nwcut_500m_lcfar3000_z0embed_fault.vtu` — the 42 358-triangle SAF surface (Physical Surface 101); this is **arg 1**.
- `safs_fault_box_nwcut_500m_lcfar3000_z0embed_bulk.vtu` — the bulk tetra mesh; auto-found by `_derive_bulk_path()` for `--write-bulk`.

The fault triangulation is verbatim from the mesher, so the on-fault geometry (and therefore the projected tractions) is fully determined upstream. See `meshing/docs/PIPELINE_raw_data_to_mesh.md` for how this mesh was built.

1.5 Stage 1 – regional stress parameters (H&Z 2004 SAFOD, 1671 m)

From `_summary.json["params"]` of the target run:

Parameter	Value	Note
`SHmax_MPa`	113.0	max horizontal principal stress
`Shmin_MPa`	49.0	min horizontal principal stress

Parameter	Value	Note
`Sv_MPa`	45.0	vertical stress at 1671 m (ρ g z)
`P_p_MPa`	16.0	hydrostatic pore pressure at 1671 m
`SHmax_azimuth_deg`	**351.0**	the only override (`--SHmax-az 351`)
`fault_strike_azimuth_hint_deg`	314.0	SAF strikes N46°W (default)
`depth_model`	`constant`	σ^0 depth-independent; all gradients = 0
`rake_sense`	right-lateral	

Because `depth_model = constant` with zero gradients, σ^0 **is the same tensor at every point** — all spatial variation in the on-fault tractions comes purely from the fault normal rotating along the curved SAF, not from depth.

1.6 Stage 2 – bulk Cauchy tensor σ^0 (`build_bulk_stress_tensor`)

Assembles the 3×3 stress in the `(east, north, up)` frame: a horizontal `[SHmax, Shmin]` block rotated by the SHmax azimuth, with `Sv` on the vertical axis. A **single source-site sign flip** converts the geological compression-negative convention to the project-internal **compression POSITIVE** (SEAS) convention. For this run, σ^0 at z = 0 (`_summary.json`):

```
         E          N         Up
E  [  50.566   -9.889     0.0  ]
N  [  -9.889  111.434     0.0  ]   MPa  (compression POSITIVE)
Up [   0.0      0.0      45.0  ]
```

(Off-diagonal −`9.889` is the SHmax/Shmin shear coupling from the 351° azimuth; the vertical row/col is decoupled with `Sv = 45`.)

1.7 Stage 3 – per-triangle fault basis (`triangle_geometry` + `build_fault_basis`)

For each of the 42 358 triangles: centroid + **unit outward normal** n̂ (`CalcOrtho`-style cross product), then the **Tandem fault basis** (CLAUDE.md "Fault-local tangent frame"):

```
s = up × n̂   (strike)      d = s × n̂   (down-dip)      n̂   (normal)
```

Strike orientation is harmonised against the **314° hint** so all `s` vectors point consistently along the SAF (sign-flips the per-triangle `s` where its azimuth disagrees with the hint by > 90°). `n_degenerate_basis = 0` for this run — every triangle yielded a valid frame.

1.8 Stage 4 – resolve traction (`resolve_traction`)

Traction vector $\mathtt{t} = \sigma^0 \cdot \hat{\mathtt{n}}$, then project onto the fault frame:

Output	Formula	Sign meaning
`sigma_n_total`	`n̂ · t`	> 0 = compression
`sigma_n_eff`	`sigma_n_total — P_p`	effective normal stress
`tau_strike`	`s · t`	**> 0 = right-lateral**
`tau_dip`	`d · t`	> 0 = down-dip
`tau_magnitude`	$\sqrt{}$`(τ_strike² + τ_dip²)`	
`rake_deg`	`atan2(τ_dip, τ_strike)`	
`mu_apparent`	`|τ| / sigma_n_eff`	apparent friction

1.9 Stage 5 – outputs

Three files in `results/safs_fault_box_nwcut_500m_lcfar3000_z0embed_hz1671_psi37_az351/`:

`…_fault_stress.vtu` (24 MB) – the target artefact

ParaView triangle surface with two data layers:

- **Cell data** (per triangle, exact projection): `sigma_n_total_MPa_cell`, `sigma_n_eff_MPa_cell`, `tau_strike_MPa_cell`, `tau_dip_MPa_cell`, plus the per-cell basis vectors `strike_vec`, `dip_vec`, `normal_vec`, `traction_vec_MPa`, and the `gmsh:physical` / `gmsh:geometrical` tags.
- **Point data** (area-weighted node averages via `cell_to_node_average`): `sigma_n_total_MPa`, `sigma_n_eff_MPa`, `tau_strike_MPa`, `tau_dip_MPa`, `tau_magnitude_MPa`, `rake_deg`, `mu_apparent`.

  Node fields are the **area-weighted average of the cell projections**, *not* a re-projection on a node-averaged basis — the two differ on a faceted fault. The verifier (Stage 6) replays this exact averaging so its point-data check matches the writer's convention.

`…_bulk_stress.vtu` (92 MB)

The 3 693 060-tet bulk mesh with cell-data `sigma_tensor_MPa` (full 6-component symmetric σ^0 at each tet centroid). For `constant` depth-model this is the same tensor everywhere.

`…_summary.json`

Parameter echo, σ^0 at $z = 0$, cell counts, and min/median/max per field.

1.10 This run's results (`_summary.json["stats"]`)

Field	min	median	max
`sigma_n_total_MPa`	46.37	73.72	112.96
`sigma_n_eff_MPa`	30.37	57.72	96.96

Field	min	median	max
tau_strike_MPa	−9.57	26.47	31.49
tau_dip_MPa	−33.94	−16.01	8.05
tau_magnitude_MPa	1.53	30.61	33.98
rake_deg	−177.92	−29.54	176.91
mu_apparent	0.016	0.562	0.641

- `fault_n_cells = 42358`, `bulk_n_cells = 3693060`, `fault_n_degenerate_basis = 0`, zero NaN vertices.
- Median μ_apparent $\approx$ 0.56 reflects $\Psi = 37^\circ$ (well-oriented, near-Byerlee): far from the $\Psi = 69^\circ$ deep-SAF "weak fault" case (μ_app $\approx$ 0.24 in `demo_safod`). The non-zero `tau_dip` median (−16 MPa) is the transitional SS↔reverse signature of the literal 1671 m magnitudes (`Sv = 45 < Shmin = 49`).

1.11 Conventions (carried end-to-end)

- Frame: UTM Zone 11 N metres, `(x = east, y = north, z = up)`, `z` $\leq$ `0` underground.
- σ^0: **compression POSITIVE** (SEAS internal) after the single source-site flip.
- `sigma_n_eff = sigma_n_total − P_p`, `P_p > 0`.
- Tandem fault basis `s = up` × `n̂`, `d = s` × `n̂`; `tau_strike > 0` ⇒ right-lateral.
- Units: **MPa** in all VTU/JSON; Pa only in the C++ `stress_safs.h5` sidecar.

(Pinned `_CONVENTION_STRING`, echoed verbatim in `_summary.json["convention"]`.)

1.12 Stage 6 – verification (optional but recommended)

`verify_onfault_stress.py` re-evaluates σ^0 analytically at **every** cell centroid and vertex and compares against the stored VTU values (full population, not a sample). For the long-named `z0embed` run, point the `--*-vtu` flags at the files in this directory:

```
conda activate pythonenv
cd stress
BASE=safs_fault_box_nwcut_500m_lcfar3000_z0embed_hz1671_psi37_az351
python code/verify_onfault_stress.py \
    --fault-vtu    results/$BASE/${BASE}_fault_stress.vtu \
    --bulk-vtu     results/$BASE/${BASE}_bulk_stress.vtu \
    --summary-json results/$BASE/${BASE}_summary.json \
    --report-json  results/$BASE/verify_report.json
```

Tolerances: 1 Pa for bulk + fault cell-data, 1e3 Pa for fault point-data. The documented 12-variant set passes 12/12 fields at machine round-off (~1e-7 Pa on a ~1e8 Pa magnitude); the `z0embed` run uses the identical writer/verifier path.

1.13 References

- `stress/README.md` — general (multi-variant, batch) pipeline reference + the 12-variant inventory.
- `stress/code/hickman_and_zoback_regional_stress_projection.py` — H&Z analytic core (`build_bulk_stress_tensor`, `fault_basis_vectors`, `resolve_traction`, `demo_safod`, the 1671 m docstrings).
- `stress/code/project_to_fault_stress.py` — the Phase 1–4 driver (CLI, `_extract_lc_tag`, writers).
- `stress/code/verify_onfault_stress.py` — Phase-8 analytic verifier.
- `stress/docs/PLAN_onfaultstress.md` (+ `_fix.md`) — accepted plan and decision trail.
- `meshing/docs/PIPELINE_raw_data_to_mesh.md` — how the `z0embed` input mesh is built.
- Hickman, S. & Zoback, M. (2004), *Stress orientations and magnitudes in the SAFOD pilot hole*, GRL — `stress/reference/`.